



Research paper

Optimal energy management of MG for cost-effective operations and battery scheduling using BWO

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ABSTRACT

The rapid growth of renewable standalone systems, with their inherent uncertainties and intermittency, poses a huge challenge to system stability. In this paper, an intelligent microgrid is proposed that fully utilizes the complementarity characteristics in PV and battery storage system (BESS), and is capable of maintaining an economic, reliable, and environmental equilibrium in daily generation scheduling. A black widow optimization (BWO) based controller is designed to determine the best possible solution under different scheduling scenarios. In addition, an exponential cost function based on the battery's degree of discharge (DOD) is introduced to calculate the battery generation cost. It will include the battery lifetime cost in the overall power generation cost. Furthermore, the cost of the diesel generator includes the CO₂ emission cost which is inversely proportional to the generator efficiency. The BWO algorithm optimized the generation cost based on these functions and increase the reliability of the grid using reliability index. The BWO algorithm finds the optimal global solution in less than 30 iterations and avoid the local optima problem. The effectiveness and convergence speed of BWO is far better than other algorithms and is validated by simulation results.

1. Introduction

Demand of electric power and its effect on the environment is a trepidation factor of this era. Conventional power grids consist of large-scale generation that consumes non-renewable sources to generate electricity, which requires high transmission costs to provide power in hilly areas. These finite non-renewable sources are depleting very quickly and require huge discourage investment for exploration. These sources are also cause the emission of greenhouse gases that are harmful for environment (Mbungu et al., 2020). Renewable sources (RESs) based microgrid (MG) provides the solution of above problem and enhance the power efficiency. Currently, photovoltaic (PV) and wind energy are the most prominent renewable sources. Generation of these RESs are nonlinear in nature due to its intermittent behavior and lead to issues related to control and reliability within MGs (Roslan et al., 2021). Modern MGs considered RESs as a primary source whereas energy storage and dispatchable sources are considered as secondary sources. MGs include energy storage and dispatchable sources with energy management system (EMS) to provide smooth and uninterrupted operation. Load following and cycle charging strategies are conventionally employed in MGs for EMS operation to balance the supply and demand (Siti et al., 2022; Hussan et al., 2024). Load flow refers to prioritize the load demand while cycle charging refers to enhance and

increase the life of the energy storage system (Sadarani et al., 2023; Jendoubi et al., 2020). The presence of nonlinear load decreases the efficiency of MGs like voltage fluctuations and severe harmonics. An effective EMS is required to optimize the operation and minimize the operational cost of MGs under different uncertainty scenarios. Voltage stability, carbon emissions, and cost of secondary sources are reported as optimization objectives for MGs (Reddy et al., 2023).

Nowadays, researchers are working on optimization techniques to achieve optimized solutions for electrical problems related to MG. Implementation of optimization techniques for MGs power system are quite difficult due to nonlinear and non-differentiable behavior of large number of constraints of objectives. Nonlinear model predictive control (MPC) based coordinated and multivariable EMS is presented in Dizqah et al. (2014) for optimal power distribution, voltage regulation and battery management of microgrid. This system approach is based on local optimization that might suffer from time-consuming and inefficient convergence. A comparative analysis of dynamic performance of AC and DC bus voltage is presented in Barelli et al. (2020) for residential MG but this study is limited to normal operating conditions of grid-connected MGs only. In Hossain et al. (2019), PSO based optimal EMS is proposed for community MG that considers battery degradation and dynamic optimal cost. A heuristic approach for analyses of optimization

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Nomenclature

ΔT	Change in Temperature
C_{bat}	Cost of the Battery Power generation
$C_{CO_2-emission}$	Cost of the CO ₂ emissions
C_{DG}	Cost of the power generated by DG
E_{ENS}	Estimated Energy not served
E_{total}	Total Energy Demand
FF	Fill Factor
G	Solar Irradiance
I_{sc}	Short Circuit Current
K	Temperature Coefficient
k_{bat}	Primary cost of the Battery Power Generation
K_{DG}	Primary Cost of the DG power Generation
LHV_{fuel}	Low heating value of the fuel
P_B	Battery Power
P_L	Load Power
P_{DG}	Diesel Generator Power
P_{PV}	Photovoltaic Power
P_{rated}	Rated Power of the Battery
STC	Standard Temperature Conditions
V_{oc}	Open Circuit Voltage

outcomes of standalone MG is presented in [El Tawil et al. \(2018\)](#), although its optimal operation appears to be limited. In [Maulik and Das \(2018\)](#), bi-objective EMS is introduced without considering experimental validity to enhance the operation of MG by improving small signal operation. MPC based hybrid control strategy with the combination of iterative learning control is presented in [Zhang et al. \(2021\)](#) for EMS to manage the dynamic power changes. In this hybrid control strategy, MPC track the reference current with iterative learning control technique to mitigate the steady state error. This study has limitations to address the transient error due to slow dynamic response ([Nkwanyana et al., 2023](#)). Optimal operation of MGs requires efficient assessment of constraints such as load demand and its forecasting, emission, battery SOC, and diesel generator efficiency. MPC based power dispatch approach is presented in [Su et al. \(2014\)](#) in terms of state of charge (SOC), size and cost for battery as secondary source. Particle swarm optimization (PSO) technique is proposed in [Wang and Singh \(2009\)](#) to find the optimal power flow solution between system stability and cost without undertaking system reliability. The authors in [Rathore and Patidar \(2021\)](#), modified PSO technique for voltage stability of MG without considering the dynamics of sources and load. Stochastic operation strategy is presented in [Hemmati et al. \(2020\)](#) for optimal MG operation with reliability cost index to calculate load interruption. However, the computational time for stochastic solutions is quite large and requires higher processing systems. Optimal energy management and economic analysis is represented in [Vahedipour-Dahraie et al. \(2019\)](#) to solve the deterministic problem with market price uncertainty based on demand response participants. This work considers the demand side management to increase the system reliability. Optimal operation and reliable configuration of small scale standalone MG is presented in [Harasis et al. \(2020\)](#) to reduce the global cost. However, very limited factors were considered in the calculation of generation cost.

Holistic power management strategy is presented in [Shan et al. \(2021\)](#) to achieve optimal operational cost and mitigate voltage deviations. Equilibrium Optimizer algorithm is proposed in [Kharrich et al. \(2021\)](#), to minimize the cost and improve the reliability. Partial shading of PV system is not considered in this work. This factor is identical necessary for optimal solution because a small change can affect the

performance of MGs. Dispatch optimization strategy is proposed in [Aziz et al. \(2022\)](#) for optimal operation and results shows the better economical and environment performance as compare to load following and cycle charging strategies. Hybrid whale optimization with pattern search algorithm is presented in [Tao et al. \(2021\)](#) to achieve the dynamic performance of MG. A review of different control and optimization techniques is highlighted in [Salehi et al. \(2022\)](#) and covered different aspects like optimal power generation, emission limitation and demand side management. A novel mixed integral linear optimization model is presented in [Karimianfard et al. \(2022\)](#) for optimal operation of large scale ESS and demand side management to maximize the profit. Cost optimization of dispatchable sources and maintaining the lifetime of sources is a serious concern for above optimization techniques. Generally, battery cost is calculated with the ratio of total cost and battery capacity. Batteries lifespan, maintenance, and operational benefits are also considerable factor in optimizing and sizing of BESS. DoD and SoC are related parameters to assess and optimize the battery cost. In addition, DG cost is calculated based on the fuel cost, output power, efficiency and duty hours whereas, CO₂ is also an important factor to assess the environmental impact and cost of generation for DG.

In this paper, power sharing algorithm for cost effective operation of standalone microgrid is proposed. The focus of this article is to minimize the energy cost, and environmental effect of electricity generation in standalone microgrids. In addition, the reliability of the system is also increased. The contribution of this paper in accordance with the existing literature is summarized as follows:

- Developing adaptive power sharing approaches to achieve the optimization goals.
- Balancing between cost effective operation and reliability of the standalone microgrids.
- Minimizing the fossil fuel consumption and power generation cost of the microgrid.
- An exponential cost function based on battery's DOD is introduced to calculate the power sharing cost of the battery.
- CO₂ emission cost is added in the diesel generator cost function along with the efficiency curve to decrease the effect of fossil fuel on environment.

2. Mathematical model

A standalone microgrid that includes PV panels, battery storage units and diesel generator is considered in this paper. The general framework and primary configuration of the system is shown in [Fig. 1](#). The power sources used in the system are mathematical modeled and incorporated in the optimization algorithm as follows:

2.1. PV system modeling

The PV array output power is effected by the intensity of the solar irradiance and the change in ambient temperature. The MPPT algorithm is implemented to get the maximum power from the PV array at time t . Electrical equivalent circuit of PV array with a series resistance ([Alramlawi and Li, 2020](#)) is used to calculate the available power at time t as described in the equation

$$P_{PV} = V_{oc} \times I_{SC} \times FF \quad (1)$$

where FF is the fill factor of the PV array, V_{oc} and I_{SC} are the open circuit voltage and short circuit current of the PV array, respectively. It can be calculated by using Eq. (2) and (3).

$$V_{oc} = \left(1 + K_{V_{oc}}^G (\log(G/G_{STC}))\right) (V_{oc_{STC}} + K_{V_{oc}}^T \Delta T) \quad (2)$$

$$I_{sc} = \frac{G}{G_{STC}} \left(I_{sc_{STC}} + K_{I_{sc}}^T \Delta T\right) \quad (3)$$

Where K is the temperature coefficient and G represents the solar irradiance. The values of these constant are given in the manufacture datasheet for standard temperature conditions (STC).

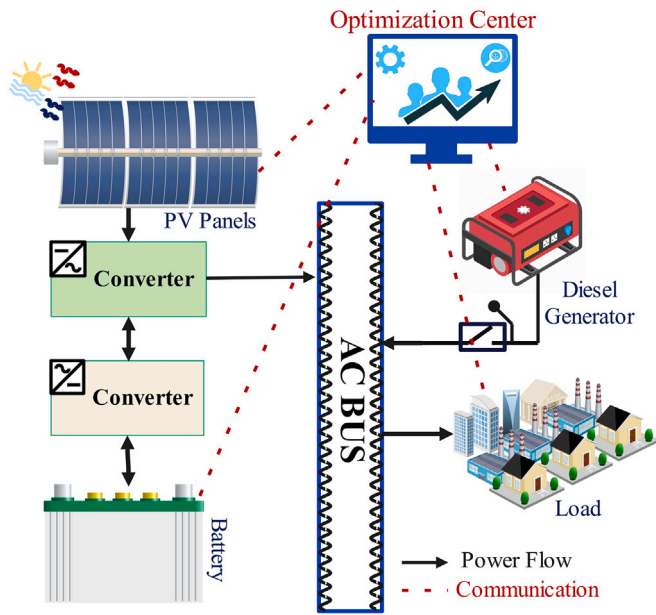


Fig. 1. Proposed system power flow diagram.

2.2. Battery storage system

The battery storage system demonstrates optimal performance when operated above a certain depth of discharge (DOD). The charging and discharging processes significantly affect the battery's lifespan and can lead to premature replacements, subsequently increasing the system's costs. In this paper, we introduced a cost function to compute the cost of battery power generation, based on depth of discharge (DOD). The cost of the power generation by battery increases exponentially with the increase in DOD value. Therefore, an early degradation factor is included in calculating the power generation cost of the batteries at time t by introducing a new variable α in the cost function represented in Eq. (4).

$$C_{bat} = k_{bat} \times e^{2\alpha} \times (P_{rated} \times SOC) \quad (4)$$

$$\alpha = 1 + DOD \quad (5)$$

Where C_{bat} is the cost of the power generated by the battery, P_{rated} is the rated power of the battery and k_{bat} is the primary power generation cost of the battery. The battery model of proposed system is shown in Fig. 2. The battery's maximum power availability is determined by using the current SOC value and its minimum SOC limit. The minimum value of SOC is also called the DOD and its optimal value is mentioned in the data sheet of the battery. Increasing the value of the DOD will result early degradation of the battery and expressed in Eq. (6).

$$\text{Battery lifetime(in cycles)} = \frac{\text{Initial capacity}}{DOD} \quad (6)$$

2.3. Reliability of the grid

As a reliability criterion, energy index reliability (EIR) is used to measures the energy deficiency level of the system (Tao et al., 2021). It can be mathematically formulated as:

$$EIR = 1 - \frac{E_{ENS}}{E_{total}} \quad (7)$$

In Eq. (7), E_{ENS} represents the estimated energy not served, while E_{total} denotes the total energy demand. As a metric for reliability assessment, the battery functions as the backup energy source. We utilize the

EIR as a metric to measure the microgrid's energy performance, aiming to attain the desired reliability level. The EIR value is determined by evaluating the ratio of the energy supplied to the total energy demand, which includes both the load requirements and the energy reserve. Furthermore, by substituting the power supplied and demanded into Eq. (7), we can express the reliability objective function as follows:

$$R(t) = \sum_{t=1}^{t=24} (P_L(t) - P_{PV}(t) - P_Z(t)) \quad (8)$$

The Eq. (9) is derived to extract the power from batteries and the diesel generator to increase the reliability of the grid.

$$P_Z = d_1 P_B + d_2 P_{DG} \quad (9)$$

d_1 and d_2 are the controllable elements which controls the power sharing between battery storage system and diesel generator. These variables are optimized to increase the reliability of the grid.

2.4. Diesel generator

The DG serves as an inertial source and possesses a slower response time when compared to inverter-based sources. Its power output demonstrates a nonlinear relationship with efficiency. Consequently, in the considered microgrid, the optimization strategy emphasizes a substantial allocation of DG power to ensure both high efficiency and economic justification. The fuel consumption of a diesel generator is contingent upon both its operation relative to its rated power and the quality of the fuel being utilized. Operating the diesel generator at 20% of its rated power results in significantly higher specific fuel consumption, corresponding to lower generator efficiency (Jakhiani et al., 2012). In contrast, the diesel generator achieves high fuel efficiency when operated at its rated power. The relationship between the electrical output and the chemical energy is depicted in Eq. (10), and the efficiency curve with respect to percentage of the rated output is shown in Fig. 3 (Energy, 2023).

$$\eta_{DG} = \frac{3.6 \times P_{DG}}{(F_o + F_1 P_{DG}) \cdot LHV_{fuel}} \quad (10)$$

where P_{DG} represents electrical output in kW, F_o and F_1 denote the fuel curve intercept coefficient and slope, respectively. LHV_{fuel} is the lower heating value of the fuel, representing its available energy content. Given the significant efficiency drop when the generator's output power falls below 30% of its rated power, it becomes imperative to enhance the penalty on the DG power sharing. As depicted in Fig. 3, power-efficiency relationship is subjected to curve fitting and subsequently incorporated into the power generation cost equation for the diesel generator (DG). This inclusion is instrumental in optimizing the objective function for calculating the grid generation cost.

$$C_{DG} = k_{DG} \frac{P_{DG}}{\eta_{DG}} + C_{CO_2-emission} \quad (11)$$

Where k_{DG} is the primary generation cost of diesel generator. CO_2 emission cost is also included in Eq. (11) to calculate the cost of electricity generated by the DG. Efficiency of diesel generator is inversely proportional to the cost and CO_2 emissions. CO_2 emission is reduced when DG is operated at rated load and its highest efficiency.

3. Problem formulation

In this paper, a multi-criteria design framework of stand-alone grid based on PV/Battery/DG system is proposed. The objectives are to minimize several costs, including those associated with components, network purchased power, CO_2 emission cost and cost of system lifespan (CSLS), while ensuring compliance with reliability constraints. The system's reliability is enhanced through power sharing and cost-effective techniques.

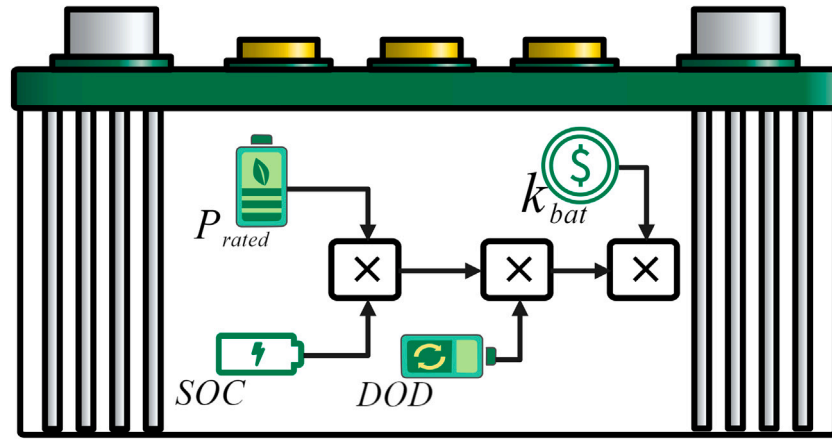


Fig. 2. Battery modeling in the proposed system.

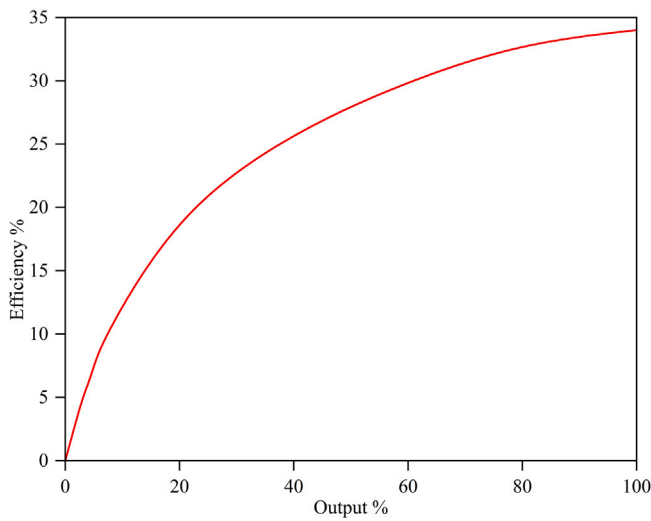


Fig. 3. Diesel generator efficiency curve.

3.1. Fitness function

The objective of this problem is to minimize the power generation cost in the microgrid and maximizing the life of the storage system and reliability of the system. The environmental effects of carbon emission are also minimized in this problem. The fitness function is designed as a multi-objective optimization using the cost function of the power sources available in the microgrid and the reliability of the microgrid as described in .

$$f(c) = \min \sum_{t=1}^{24} C_B(t) + C_{DG}(t) + R(t) \quad (12)$$

$$f(r) = \max \sum_{t=1}^{24} R(t)$$

The cost function of the battery storage system incorporated the deterioration cost in the form of DOD while the diesel generator equation has the carbon emission penalty cost in it. Therefore, the fitness function for the whole problem is elaborated in Eq. (13).

$$fitness = w_1 \times f(c) + w_2 \times f(r) \quad (13)$$

$$w_1 + w_2 = 1$$

where w_1 and w_2 are the weights given to each optimization function. The weights are adjusted according to the requirements of the microgrid.

3.2. Constraints

3.2.1. Reliability constraint

Grid reliability is a paramount concern in electrical power systems, with profound implications for both economic and societal well-being. Minimizing the operational cost and maximizing the life the system cannot be pursued at the expense of grid reliability. It is the core of our fitness function and ensure a dependable power supply. The reliability constraint for the proposed system is described in Eq. (8).

3.2.2. CO₂ Emission constraint

Our optimization problem revolves around the design and configuration of power systems, with objectives that may encompass cost minimization and performance enhancement. However, we firmly acknowledge that these objectives must not come at the expense of exacerbating the climate crisis. Mitigating carbon dioxide (CO₂) emissions is a global imperative as we grapple with the urgent challenge of climate change. In this paper, the CO₂ emission cost is included with the power generation cost of the DG.

3.2.3. SOH of the battery constraint

The State of Health (SOH) of the battery storage system is an additional constraint that is considered when formulating the fitness function. This inclusion is essential to prolonging the overall lifespan of the batteries. If batteries degrade prematurely, it can lead to reduced energy storage capacity, shorter backup times, effects the reliability and increased operational costs. Furthermore, by preventing premature battery failure, the microgrid can contribute to sustainability goals by reducing the environmental impact associated with battery production and disposal. The SOH function of the battery is included in the objective function of the system.

4. Black widow optimization algorithm

In this paper, we propose the utilization of the Black Widow Optimization Algorithm (BWO) as a population-based optimization technique to optimize the power flow within the microgrid. Inspired by the lifecycle and unique mating behavior of black widow spiders, the BWO algorithm offers rapid convergence and effectively mitigates the local optima problem. Its capability to explore a wide search space ensures the identification of the optimal global solution, making it well-suited for addressing microgrid optimization challenges characterized by multiple local optima.

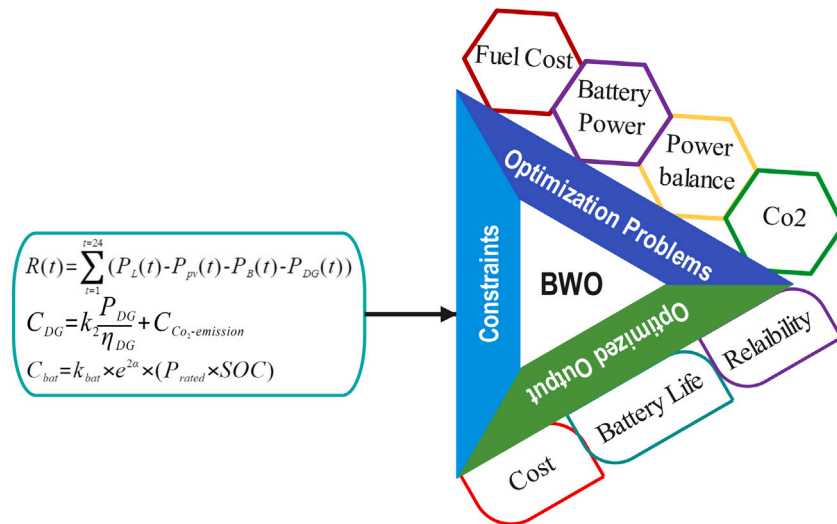


Fig. 4. BWO optimization.

To enhance the effectiveness of the BWO algorithm in obtaining superior solutions, careful adjustment of certain parameters is crucial. These parameters, including the procreation rate (PP), cannibalism rate (CR), and mutation rate (PM), play vital roles in guiding the algorithm towards optimal outcomes. Adjusting these parameters appropriately can significantly improve the algorithm's performance and its ability to find high-quality solutions. Proper adjustment of these parameters is essential to enhance the algorithm's effectiveness in discovering superior solutions. Cannibalism is selected by pairing the parameter values with other parameters. The parameter which fits in more than one situation will survive other will die out. Some parameters should be declared primary and other are secondary. Secondary parameters are selected based on their combination with primary parameters. If changing a parameter necessitates adjusting numerous other parameters, it may be omitted. The implementation of BWO in the proposed work is depicted in Fig. 4.

The BWO optimization process begins by generating a population of spiders, each spider representing a random solution for power-sharing, cost minimization, reliability, and state-of-health (SOH) of the battery. Subsequently, these solutions are utilized to optimized the system variables at each iteration. These variables include the power-sharing among different sources, the power demand, the SOH of the battery, and the generation cost. Next, the calculated cost is fed back into the optimization process. Iteration after iteration, and with consideration of the number of particles, the positions of the spiders are updated until the optimization converges, reaching the global optimal solution.

$$window = \{s_1, s_2, \dots, s_{n-p}\}, \quad s = \begin{Bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{n-p} \end{Bmatrix} \quad (14)$$

$$Fitness = f(window) = f(s_1, s_2, \dots, s_n) \quad (15)$$

In Eq. (14) spiders are represented by s , and the population of spiders is denoted by $n - p$. Widow function contains $n - p$ number of spiders and each spiders contain v_{n-p} parameters. Fitness function in Eq. (15) contains n possible solutions of the problem and arrange them in the descending order of their fitness value. It will check the fitness value of each spider s on each iteration and arrange them accordingly. The process and hierarchy of BWO is shown using flowchart in Fig. 5.

5. Results and discussion

Simulation based method is used to analyze the performance of microgrid with BWO algorithm. The simulation runs using the MATLAB

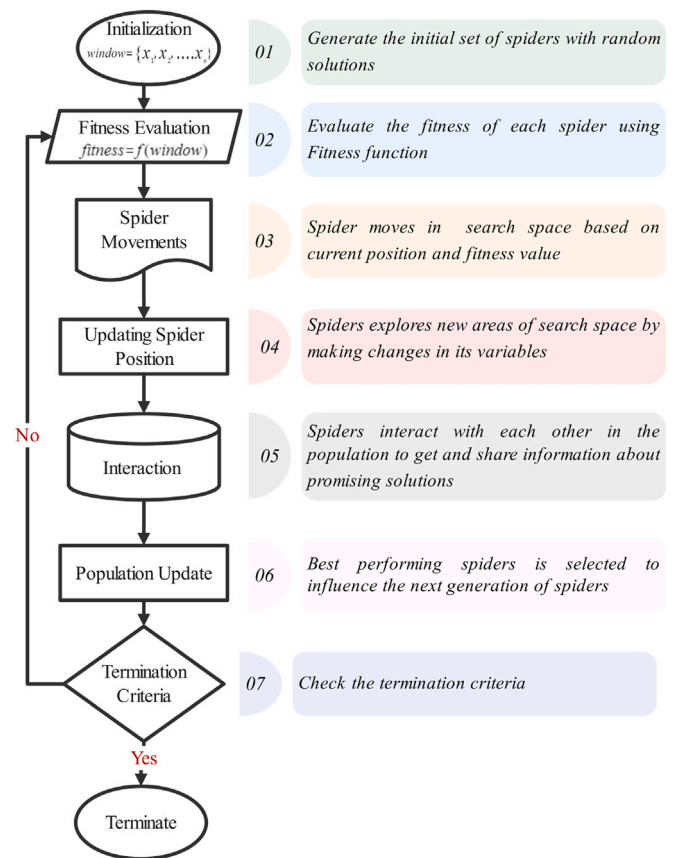


Fig. 5. Black widow optimization algorithm flow chart.

2021a software on a CoreI7 system having Nvidia RTX3050 GPU. The performance of proposed algorithm is tested from two aspects: prolonged analysis in days and transient analysis in second. Prolonged analysis helps to show the energy management and economical operation whereas, transient analysis helps to deeply analysis the battery health and voltage profile. Real time load profile data of the houses in US extracted from the dataset (Kolter and Johnson, 2011) to analyze the performance of the proposed algorithm. Load profile of houses for 72 h is shown in Fig. 6. PV power generation data is extracted from the

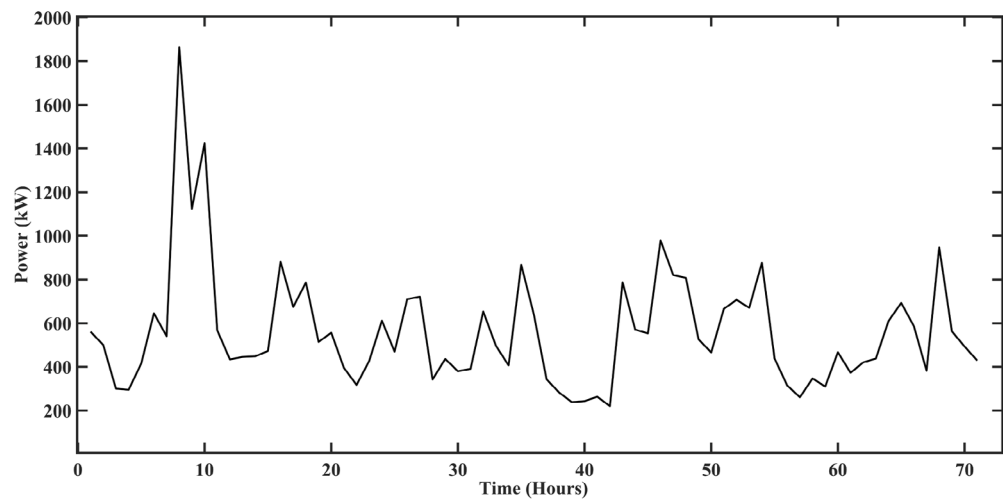


Fig. 6. Realtime load curve for 72 h.

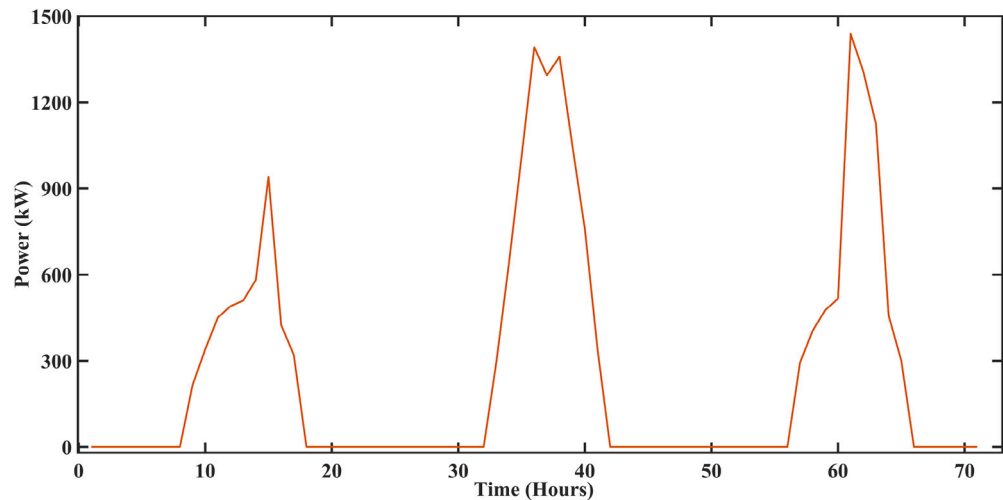


Fig. 7. PV power.

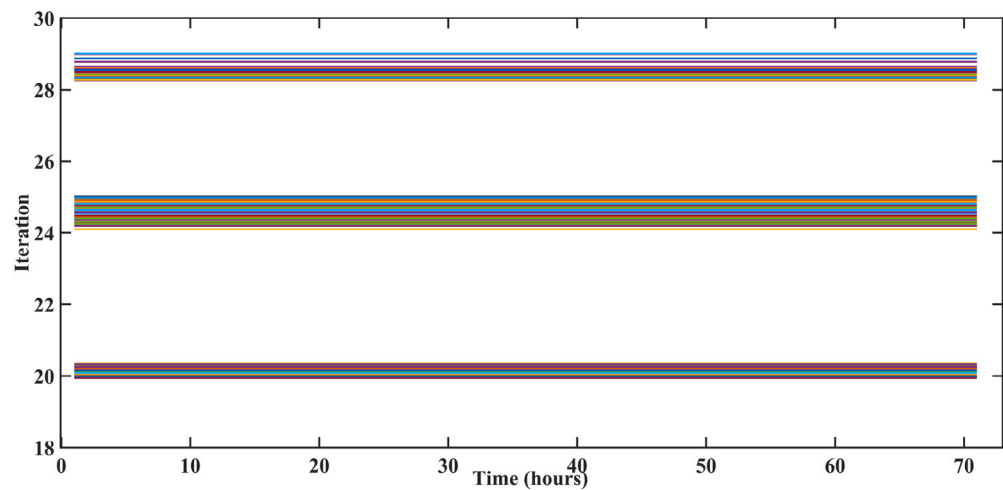


Fig. 8. BWO spiders.

NREL database for the Massachusetts state (NREL, 2023) and is shown in Fig. 7. The PV system is designed to give 1400 kW power at STC. The average PV generation of day 1–3 are 178 kW, 400 kW and 263.5 kW

respectively. Optimal microgrid operation is difficult task due to big difference between generation and demand that encouraged to select this data for testing. The primary source of generation is PV system

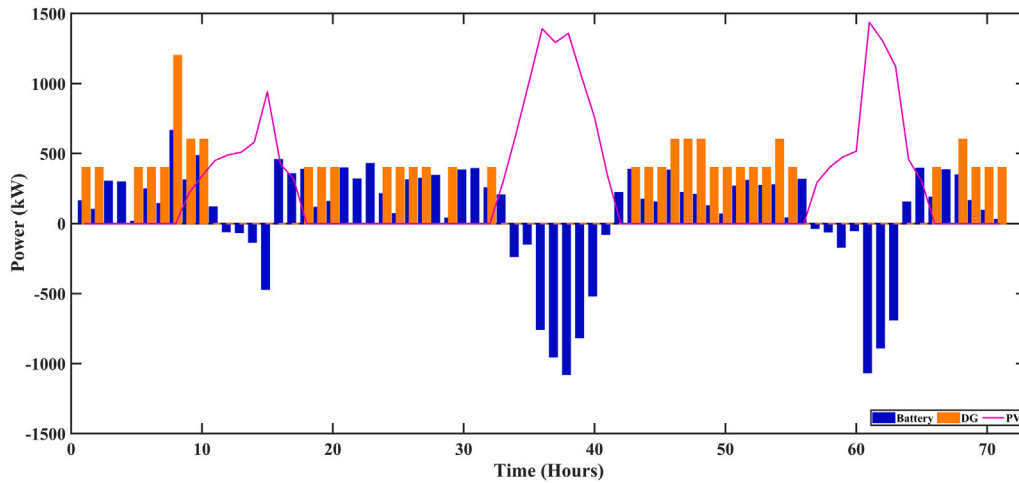


Fig. 9. Power sharing curves between sources.

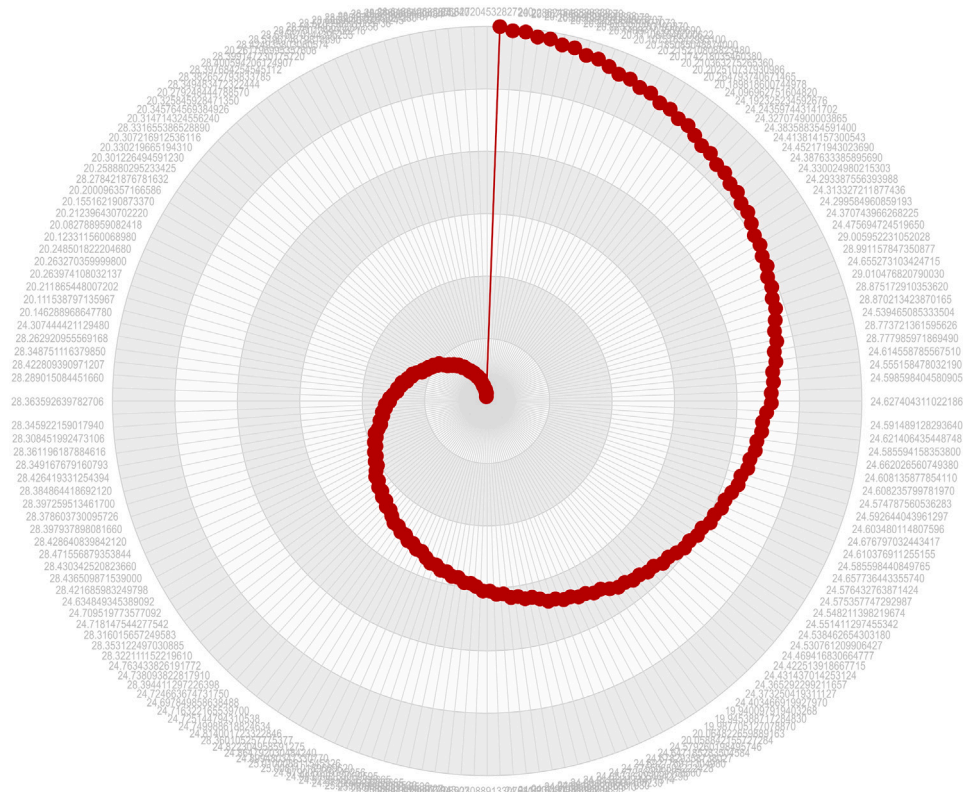


Fig. 10. Searching the optimal solutions for resource allocation.

whereas the secondary source of energy sources are battery and diesel generator. BWO ensure the optimal utilization of all available energy to fulfill the load demand by optimal resource allocation. When the microgrid have excess PV power then surplus power will provide to battery storage system and if PV have less power than load demand then secondary sources will provide the power. The power contribution at high load demand and low PV generation requires secondary sources. Battery power contribution will depend on its health that will calculate from Eq. (6). If the battery lifetime cost is less than diesel generator power cost, then battery will be the optimal choice to supply the power. Furthermore, battery charging will be prioritized when DG operates at below its rated power to increase the efficiency and decrease the CO₂

emissions. The optimal power sharing combination is chosen by the BWO algorithm and is called as best spider as shown in Fig. 8. The BWO assigns the fitness values to all spiders at each iteration and achieve the best fitness value during the optimization process. The optimal power sharing between PV, batteries and diesel generator to fulfill the load demand is shown in Fig. 9. It will show that the minimum operating point of the diesel generator is 400 kW. The BWO will turn ON the diesel generator whenever there is a shortage of power more than 400 kW in the system. This will help in maintaining a better efficiency of the DG and reducing its fuel consumption. The iterative path taken by the BWO algorithm to achieve the global optimal point is shown in Fig. 10. The BWO converges to the global optima quickly by taking least

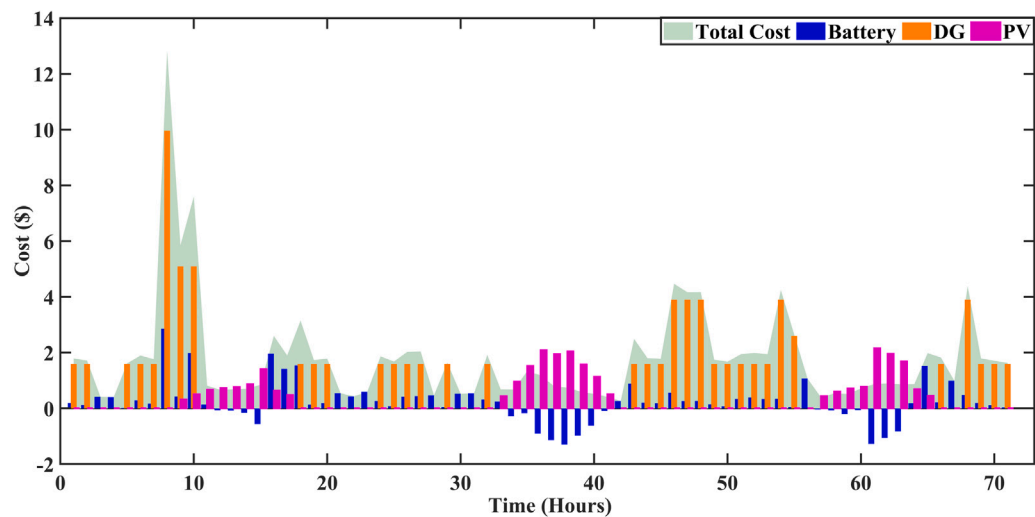


Fig. 11. Per unit cost.

Table 1
Cost factors.

Factor	k_{bat}	k_{pv}	k_{DG}
Value	1.2 kWh	1.5 kWh	4 kWh

number of iterations. It will swiftly identify cost-effective solutions for the microgrid, outperforming other algorithms in terms of speed. The BWO allocate the power to energy sources for optimal cost solution by considering the available power, battery health, environmental effect and grid reliability. If battery contribution will affect the battery health, then battery cost will increase and if the battery deterioration cost increases the cost of diesel generator, the algorithm turn on the diesel generator. By considering the cost factors mentioned in Table 1, the per unit cost and the cost optimization solution achieved by the proposed algorithm are shown in Fig. 11 and Fig. 12 respectively.

The superiority of proposed BWO algorithm is tested against the performance of Gray Wolf Optimization and Particle Swarm Optimization techniques. Average fitness value for find the optimal solution with 100 iterations using BWO, PSO and BWO is shown in Fig. 13. Average fitness value helps the measure how close the algorithm is to finding an optimal solution. Results clearly shows BWO achieve fitness value in least number of iterations as compare to GWO and PSO. The proposed system optimizes the load sharing curves in the run time and no forecasted data is stored in the system. It only stores the runtime data of power generation sources available in the microgrid. Therefore, transient analysis of the proposed algorithm is necessary and is shown in Fig. 14. The proposed system performance is observed for 18 sec data of load and PV for transient analysis. Load and the generation of PV are not equal and continues changing. Therefore, fast and optimal battery backup operation is required to ensure the optimal operation. When the load demand is greater than the generation, battery will provide backup power and when the load demand is less than generation then battery consume the excess power for charging operation. The transient analysis ensures the reliability of the system. The proposed BWO algorithm maintains better voltage regulation and still proved optimal power sharing solutions. The output power of the battery that ensures the reference battery current and output voltage tracking is shown in Fig. 15.

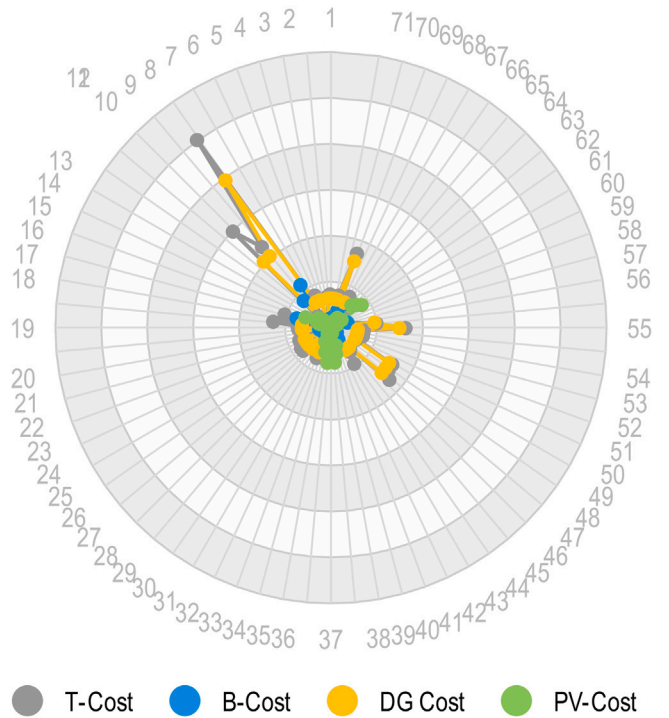


Fig. 12. Optimal solutions for cost optimization.

6. Discussion

The proposed work enables an efficient operation of standalone microgrid. The BWO optimizes the cost of generation and keeping the CO₂ emission at its minimum. Furthermore, it will keep the diesel generator running at its maximum efficiency which helps in reducing the fuel consumption and mitigating its environmental impact. In addition, the charging and discharging of the battery storage system is not compromised by the reliability of the grid. This will increase the system life and reduce the overall operational cost of the system. The prolong and transient analysis of the system under real time load

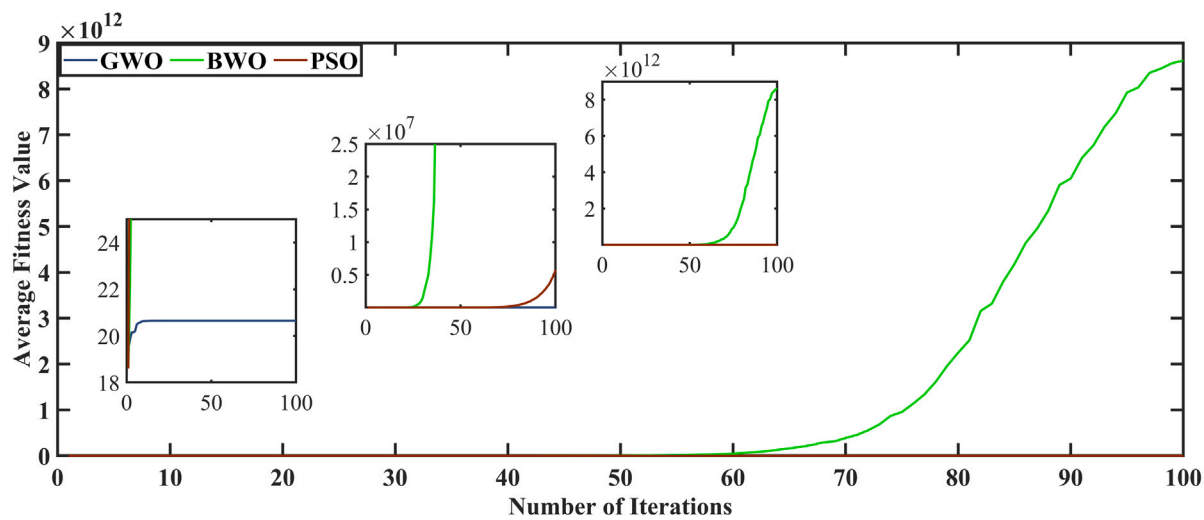


Fig. 13. Comparison.

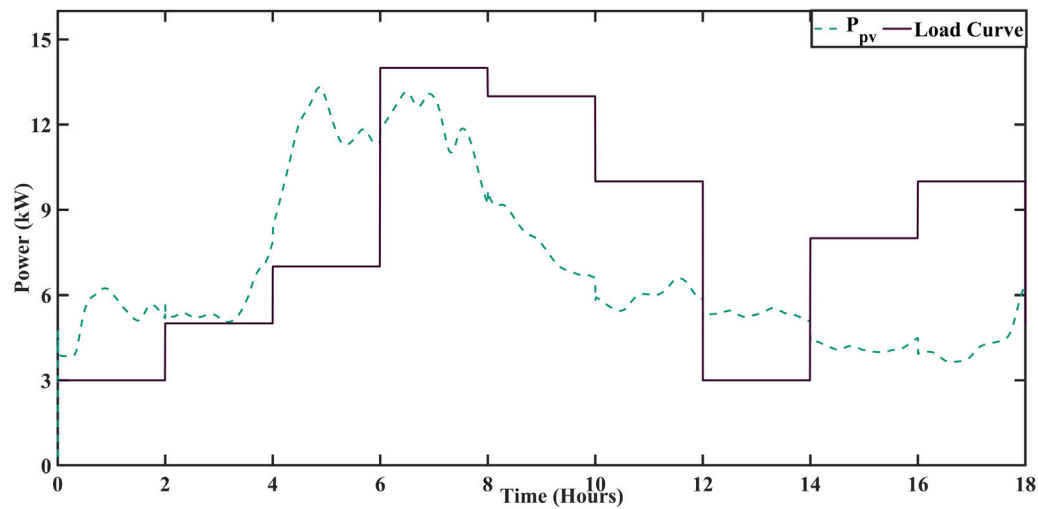


Fig. 14. Load and PV generation for 18 s.

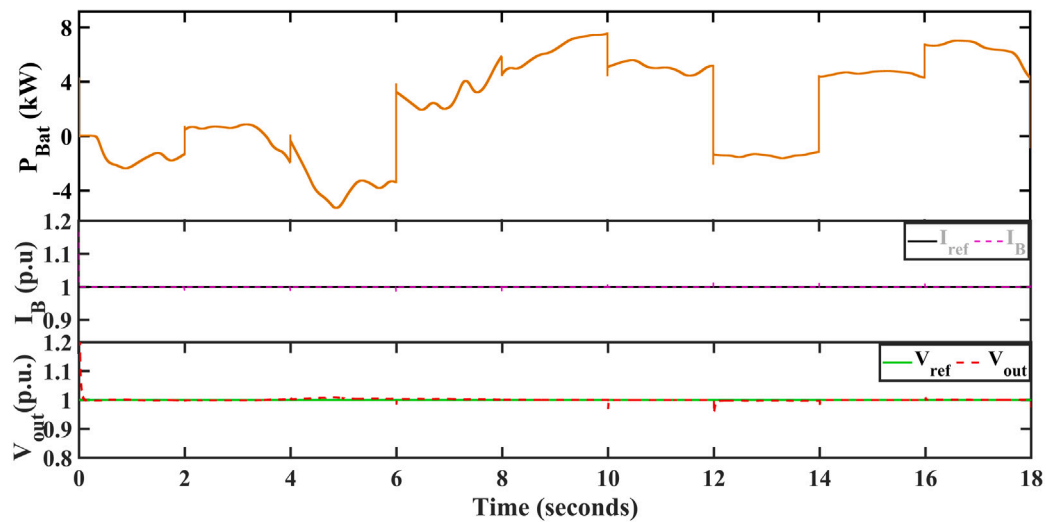


Fig. 15. Battery contribution, current tracking and output voltage.

profile shows that BWO based grid optimization behaves well under all circumstances.

7. Conclusion

This manuscript presents a novel BWO-based microgrid energy management algorithm that offers significant advancements in cost-effective operation, battery scheduling, and environmental sustainability. By leveraging the complementarity characteristics of PV and battery storage system, the algorithm ensures economic, reliable, and environmentally sustainable generation scheduling in microgrid operations. Through the integration of a BWO based controller, the algorithm can efficiently determine the best solution across various scheduling scenarios, thereby enhancing the grid's overall performance.

The contribution of this research lies in the development of adaptive power-sharing approaches tailored to achieve optimization objectives while balancing cost-effective operations and system reliability. By minimizing fossil fuel consumption and power generation costs, the algorithm significantly reduces the environmental impact of electricity generation in standalone microgrids. Additionally, the incorporation of an exponential cost function based on the battery's DOD enables accurate calculation of the battery generation cost, including considerations for battery lifetime costs. Moreover, the inclusion of CO₂ emission costs in the diesel generator cost function, coupled with efficiency considerations, serves to mitigate the environmental footprint of fossil fuel usage in microgrid operations.

Overall, the findings presented in this manuscript underscore the effectiveness and efficiency of the proposed algorithm in optimizing energy management, reducing costs, and enhancing the reliability of standalone microgrid systems. Through rigorous simulation and validation, the algorithm demonstrates superior performance compared to existing approaches, thus offering promising prospects for real-world implementation and deployment in microgrid applications.

CRedit authorship contribution statement

Muhammad Ahsan Ayub: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Umair Hussan:** Writing – original draft, Visualization, Methodology, Investigation. **Hamna Rasheed:** Writing – review & editing, Investigation, Conceptualization. **Yitao Liu:** Writing – review & editing, Validation, Supervision, Formal analysis. **JianChun Peng:** Writing – review & editing, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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